

Universidad del Istmo

Facultad de Ingeniería

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Arquitectura de computadoras y microcontroladores

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Arquitectura de computadoras y microcontroladores

P3

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Ingeniería en Sistemas y Ciencias de la Comunicación

Parte II

Saif en vivado con comando TCL : restart
open_saif "alu_secuencial.saif"
log_saif [get_objects -r /testbench_Top_ALU_secuencial/uut/*]
run all
close_saif

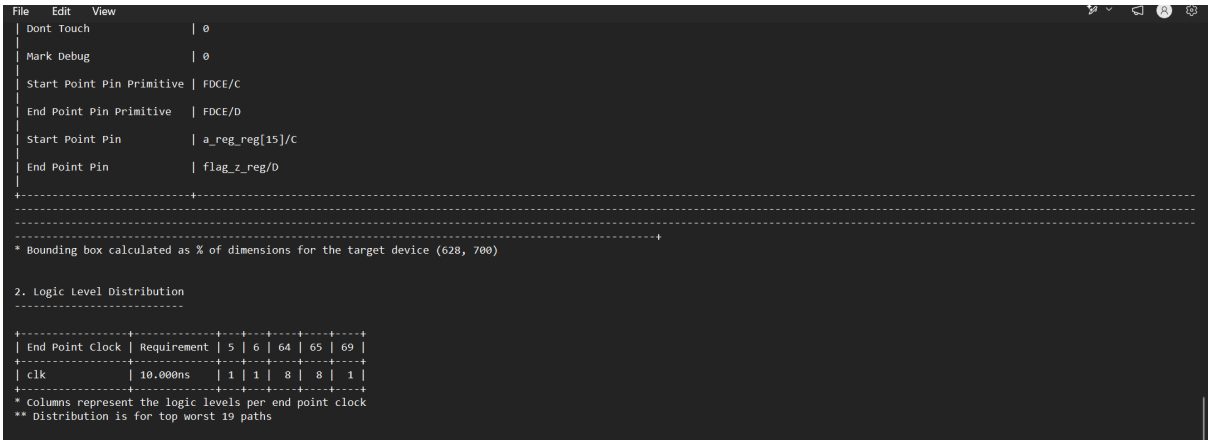
Restricciones SDC para memoria :
create_clock -period 10.000 -name clk [get_ports clk] set_clock_uncertainty 0.200 [get_clocks clk] set_input_delay -clock [get_clocks clk] 2.000 [remove_from_collection [all_inputs] [get_ports clk]] set_output_delay -clock [get_clocks clk] 2.000 [all_outputs]

Saif :

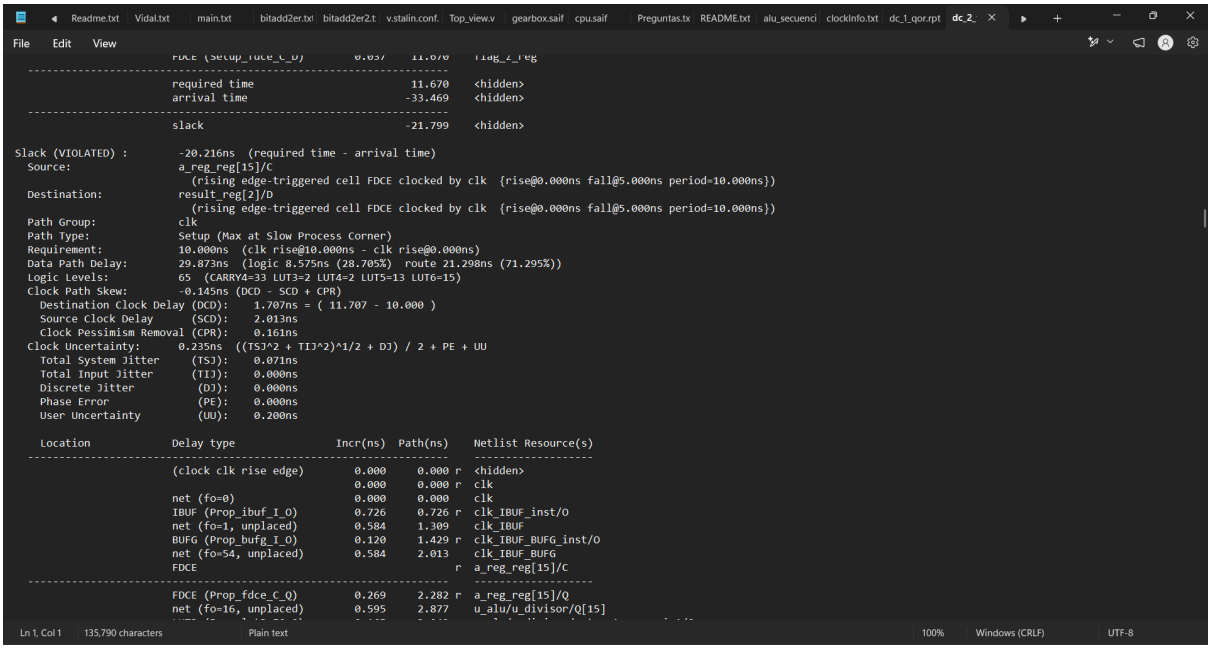
```
(SAIFILE
   (SAIFVERSION "2.0")
   (DIRECTION "backward")
   (DESIGN )
   (DATE "Mon Oct 19 21:21:32 2026")
   (VENDOR "Xilinx, Inc")
   (PROGRAM_NAME "Vivado Simulator")
   (VERSION "2020.2")
   (DIVIDER /)
   (TIMESCALE  1 ps)
   (DURATION  1055000)
   (INSTANCE  testbench_Top_ALU_secuencial
      (INSTANCE  uut
         (NET
            (clk (T0 530000) (T1 525000) (TX 0) (TZ 0) (TB 0) (TC 210))
            (rst_n (T0 15000) (T1 1040000) (TX 0) (TZ 0) (TB 0) (TC 1))
            (a\[0\] (T0 575000) (T1 480000) (TX 0) (TZ 0) (TB 0) (TC 36))
            (a\[1\] (T0 580000) (T1 475000) (TX 0) (TZ 0) (TB 0) (TC 37))
            (a\[2\] (T0 540000) (T1 515000) (TX 0) (TZ 0) (TB 0) (TC 35))
            (a\[3\] (T0 420000) (T1 635000) (TX 0) (TZ 0) (TB 0) (TC 37))
            (a\[4\] (T0 400000) (T1 655000) (TX 0) (TZ 0) (TB 0) (TC 33))
            (a\[5\] (T0 405000) (T1 650000) (TX 0) (TZ 0) (TB 0) (TC 28))
            (a\[6\] (T0 485000) (T1 570000) (TX 0) (TZ 0) (TB 0) (TC 44))
            (a\[7\] (T0 375000) (T1 680000) (TX 0) (TZ 0) (TB 0) (TC 36))
            (a\[8\] (T0 500000) (T1 555000) (TX 0) (TZ 0) (TB 0) (TC 35))
            (a\[9\] (T0 495000) (T1 560000) (TX 0) (TZ 0) (TB 0) (TC 42))
            (a\[10\] (T0 625000) (T1 430000) (TX 0) (TZ 0) (TB 0) (TC 42))
            (a\[11\] (T0 545000) (T1 510000) (TX 0) (TZ 0) (TB 0) (TC 42))
            (a\[12\] (T0 580000) (T1 475000) (TX 0) (TZ 0) (TB 0) (TC 37))
            (a\[13\] (T0 560000) (T1 495000) (TX 0) (TZ 0) (TB 0) (TC 43))
            (a\[14\] (T0 525000) (T1 530000) (TX 0) (TZ 0) (TB 0) (TC 36))
            (a\[15\] (T0 555000) (T1 500000) (TX 0) (TZ 0) (TB 0) (TC 50))
            (b\[0\] (T0 450000) (T1 605000) (TX 0) (TZ 0) (TB 0) (TC 45))
            (b\[1\] (T0 725000) (T1 330000) (TX 0) (TZ 0) (TB 0) (TC 34))
            (b\[2\] (T0 575000) (T1 480000) (TX 0) (TZ 0) (TB 0) (TC 30))
            (b\[3\] (T0 580000) (T1 475000) (TX 0) (TZ 0) (TB 0) (TC 31))
            (b\[4\] (T0 470000) (T1 585000) (TX 0) (TZ 0) (TB 0) (TC 47))
            (b\[5\] (T0 650000) (T1 405000) (TX 0) (TZ 0) (TB 0) (TC 39))
            (b\[6\] (T0 570000) (T1 485000) (TX 0) (TZ 0) (TB 0) (TC 43))
            (b\[7\] (T0 555000) (T1 500000) (TX 0) (TZ 0) (TB 0) (TC 40))
```

1. **Preparación:** instancie la ALU+ de 16 bits e incorpore registros de entrada y salida para implementar un sistema síncrono. Verifique su funcionamiento con VCS y Verdi.
2. **Síntesis — DC shell:** configure las bibliotecas y restricciones SDC según lo visto en clase. Genere la netlist y los reportes de área, celdas y timing. Identifique y explique la ruta crítica.
3. **Diseño físico — ICC2:** complete floorplan, alimentación, placement, síntesis del árbol de reloj (CTS) y routing. Revise y corrija las violaciones temporales y físicas reportadas.
4. **Resultados:** exporte el GDSII y compare área y timing después de síntesis y routing. Explique cómo se transformaron los bloques del RTL, incluyendo el tratamiento de los buses triestado.
5. **Entregables:** RTL, testbench, scripts Tcl, SDC, netlists, base de datos final de ICC2, GDSII y un informe breve con reportes y capturas de las principales etapas.
6. **Presentación — 15 minutos:** demuestre el funcionamiento y explique cómo su descripción HDL se convirtió en celdas físicas interconectadas. Justifique las decisiones tomadas y los resultados obtenidos.
7. **Discusion:**
    1. Generar y analizar los siguientes reportes para DC Shell:

report_qor



report_timing



```
                        required time                         11.670    <hidden>
                        arrival time                         -33.469    <hidden>
  ------------------------------------------------------------------
                        slack                                -21.799    <hidden>

Slack (VIOLATED) :        -20.216ns  (required time - arrival time)
  Source:                 a_reg_reg[15]/C
                            (rising edge-triggered cell FDCE clocked by clk  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:            result_reg[2]/D
                            (rising edge-triggered cell FDCE clocked by clk  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:             clk
  Path Type:              Setup (Max at Slow Process Corner)
  Requirement:            10.000ns  (clk rise@10.000ns - clk rise@0.000ns)
  Data Path Delay:        29.873ns  (logic 8.575ns (28.705%)  route 21.298ns (71.295%))
  Logic Levels:           65  (CARRY4=33 LUT3=2 LUT4=2 LUT5=13 LUT6=15)
  Clock Path Skew:        -0.145ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):    1.707ns = ( 11.707 - 10.000 )
    Source Clock Delay      (SCD):    2.013ns
    Clock Pessimism Removal (CPR):    0.161ns
  Clock Uncertainty:      0.235ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE + UU
    Total System Jitter     (TSJ):    0.071ns
    Total Input Jitter      (TIJ):    0.000ns
    Discrete Jitter          (DJ):    0.000ns
    Phase Error              (PE):    0.000ns
    User Uncertainty         (UU):    0.200ns

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
                         (clock clk rise edge)        0.000     0.000 r  <hidden>
                                                      0.000     0.000 r  clk
                         net (fo=0)                   0.000     0.000    clk
                         IBUF (Prop_ibuf_I_O)         0.726     0.726 r  clk_IBUF_inst/O
                         net (fo=1, unplaced)         0.584     1.309    clk_IBUF
                         BUFG (Prop_bufg_I_O)         0.120     1.429 r  clk_IBUF_BUFG_inst/O
                         net (fo=54, unplaced)        0.584     2.013    clk_IBUF_BUFG
                         FDCE                                       r  a_reg_reg[15]/C
  -------------------------------------------------------------------    -------------------
                         FDCE (Prop_fdce_C_Q)         0.269     2.282 r  a_reg_reg[15]/Q
                         net (fo=16, unplaced)        0.595     2.877    u_alu/u_divisor/Q[15]
```

report_area

```
Copyright 1986-2022 Xilinx, Inc. All Rights Reserved. Copyright 2022-2026 Advanced Micro Devices, Inc. All Rights Reserved.
----------------------------------------------------------------------------------------------------------------------------------------------------------------
| Tool Version : Vivado v.2026.1 (win64) Build 6511674 Tue Jun 16 11:02:23 MDT 2026
| Date         : Mon Oct 19 21:33:21 2026
| Host         : Tzarina12 running 64-bit major release  (build 9200)
| Command      : report_utilization -hierarchical -file dc_3_area_hierarchy.rpt
| Design       : Top_ALU_secuencial
| Device       : xc7vx485tffg1157-1
| Speed File   : -1
| Design State : Synthesized
----------------------------------------------------------------------------------------------------------------------------------------------------------------

Utilization Design Information

Table of Contents
-----------------
1. Utilization by Hierarchy

1. Utilization by Hierarchy
---------------------------

+-------------------------+---------+------------+------------+---------+------+-----+--------+--------+------------+
|         Instance        |  Module | Total LUTs | Logic LUTs | LUTRAMs | SRLs | FFs | RAMB36 | RAMB18 | DSP Blocks |
+-------------------------+---------+------------+------------+---------+------+-----+--------+--------+------------+
| Top_ALU_secuencial      |   (top) |        709 |        709 |       0 |    0 |  54 |      0 |      0 |          0 |
|   (Top_ALU_secuencial)  |   (top) |         10 |         10 |       0 |    0 |  54 |      0 |      0 |          0 |
|   u_alu                 | Top_ALU |        699 |        699 |       0 |    0 |   0 |      0 |      0 |          0 |
|     u_divisor           | divisor |        699 |        699 |       0 |    0 |   0 |      0 |      0 |          0 |
+-------------------------+---------+------------+------------+---------+------+-----+--------+--------+------------+
```

report_power

2.1 Environment

Ambient Temp (C)	25.0
ThetaJA (C/W)	1.4
Airflow (LFM)	250
Heat Sink	medium (Medium Profile)
ThetaSA (C/W)	2.4
Board Selection	medium (10"x10")
# of Board Layers	12to15 (12 to 15 Layers)
Board Temperature (C)	25.0

2.2 Clock Constraints

Clock	Domain	Constraint (ns)
clk	clk	10.0

3. Detailed Reports

3.1 By Hierarchy

Name	Power (W)
Top_ALU_secuencial	0.019
u_alu	0.007
u_divisor	0.007

Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)	Powerup (A)	Budget (A)	Margin (A)
Vccint	1.000	0.145	0.011	0.134	NA	Unspecified	NA
Vccaux	1.800	0.038	0.001	0.038	NA	Unspecified	NA
Vcco33	3.300	0.000	0.000	0.000	NA	Unspecified	NA
Vcco25	2.500	0.000	0.000	0.000	NA	Unspecified	NA
Vcco18	1.800	0.005	0.004	0.001	NA	Unspecified	NA
Vcco15	1.500	0.000	0.000	0.000	NA	Unspecified	NA
Vcco135	1.350	0.000	0.000	0.000	NA	Unspecified	NA
Vcco12	1.200	0.000	0.000	0.000	NA	Unspecified	NA
Vccaux_io	1.800	0.000	0.000	0.000	NA	Unspecified	NA
Vccbram	1.000	0.003	0.000	0.003	NA	Unspecified	NA
MGTAVcc	1.000	0.000	0.000	0.000	NA	Unspecified	NA
MGTAVtt	1.200	0.000	0.000	0.000	NA	Unspecified	NA
MGTVccaux	1.800	0.000	0.000	0.000	NA	Unspecified	NA
MGTZVccl	1.075	0.000	0.000	0.000	NA	Unspecified	NA
MGTZAVcc	1.075	0.000	0.000	0.000	NA	Unspecified	NA
MGTZVcch	1.800	0.000	0.000	0.000	NA	Unspecified	NA
Vccadc	1.800	0.020	0.000	0.020	NA	Unspecified	NA

1. Summary

Total On-Chip Power (W)	0.261
Design Power Budget (W)	Unspecified*
Power Budget Margin (W)	NA
Dynamic (W)	0.019
Device Static (W)	0.243
Effective TJA (C/W)	1.4
Max Ambient (C)	84.6
Junction Temperature (C)	25.4
Confidence Level	Low
Setting File	---
Simulation Activity File	---
Design Nets Matched	NA

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

On-Chip	Power (W)	Used	Available	Utilization (%)
Clocks	0.002	3	---	---
Slice Logic	0.005	1052	---	---
LUT as Logic	0.004	709	303600	0.23
CARRY4	<0.001	96	75900	0.13
Register	<0.001	54	607200	<0.01
Others	0.000	3	---	---
Signals	0.004	798	---	---
I/O	0.008	57	600	9.50
Static Power	0.243			
Total	0.261			

report_constraint -all_violators

```
Copyright 1986-2022 Xilinx, Inc. All Rights Reserved. Copyright 2022-2026 Advanced Micro Devices, Inc. All Rights Reserved.
------------------------------------------------------------------------------------------------------------------------------------------------------
| Tool Version : Vivado v.2026.1 (win64) Build 6511674 Tue Jun 16 11:02:23 MDT 2026
| Date         : Mon Oct 19 21:33:25 2026
| Host         : Tzarina12 running 64-bit major release  (build 9200)
| Command      : report_drc -file dc_5_violators.rpt
| Design       : Top_ALU_secuencial
| Device       : xc7vx485tffg1157-1
| Speed File   : -1
| Design State : Synthesized
------------------------------------------------------------------------------------------------------------------------------------------------------

Report DRC

Table of Contents
-----------------
1. REPORT SUMMARY
2. REPORT DETAILS

1. REPORT SUMMARY
-----------------
            Netlist: netlist
          Floorplan: synth_1
      Design limits: <entire design considered>
           Ruledeck: default
             Max checks: <unlimited>
             Checks found: 3
+----------+------------------+-----------------------------------------------------+--------+
| Rule     | Severity         | Description                                         | Checks |
+----------+------------------+-----------------------------------------------------+--------+
| NSTD-1   | Critical Warning | Unspecified I/O Standard                            | 1      |
| UCIO-1   | Critical Warning | Unconstrained Logical Port                          | 1      |
| CFGBVS-1 | Warning          | Missing CFGBVS and CONFIG_VOLTAGE Design Properties | 1      |
+----------+------------------+-----------------------------------------------------+--------+
```

Evaluación Fisica ICC2 usando Vivado:

** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more i

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
0	Yes	-	Set
55	Yes	-	Reset
0	Yes	Set	-
0	Yes	Reset	-

2. Slice Logic Distribution

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice	211	0	0	75900	0.28
SLICEL	121	0			
SLICEM	90	0			
LUT as Logic	724	0	0	303600	0.24
using O5 output only	0				
using O6 output only	549				
using O5 and O6	175				
LUT as Memory	0	0	0	130800	0.00
LUT as Distributed RAM	0	0			
using O5 output only	0				
using O6 output only	0				
using O5 and O6	0				
LUT as Shift Register	0	0			
using O5 output only	0				
using O6 output only	0				
using O5 and O6	0				
Slice Registers	55	0	0	607200	<0.01
Register driven from within the Slice	12				
Register driven from outside the Slice	43				
LUT in front of the register is unused	9				
LUT in front of the register is used	34				

| Design Timing Summary

WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints	TPWS Total Endpoints
-17.367	-298.006	18	20	0.074	0.000	0	20	4.600	0.000	0	56

Timing constraints are not met.

| Clock Summary

Clock	Waveform(ns)	Period(ns)	Frequency(MHz)
clk	{0.000 5.000}	10.000	100.000

| Intra Clock Table

Clock	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints	TPWS Total Endpoints
clk	-17.367	-298.006	18	20	0.074	0.000	0	20	4.600	0.000	0	56

4. Clock Regions: Key Resource Utilization

Clock Region Name	Global Clock Used	Global Clock Avail	BUFRs Used	BUFRs Avail	BUFMRs Used	BUFMRs Avail	BUFIOs Used	BUFIOs Avail	MMCM Used	MMCM Avail	PLL Used	PLL Avail	GT Used	GT Avail	PCI Used	PCI Avail	ILOGIC Used	ILOGIC Avail	OLOGIC Used	OLOGIC Avail	FF Used	FF Avail	LUTM Used	LUTM Avail	RAMB18 Used	RAMB18 Avail	RAMB36 Used	RAMB36 Avail	DSP48E2 Used	DSP48E2 Avail
X0Y0	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y0	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y1	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	4900	0	2200	0	140	0	70	0	180
X1Y1	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y2	1	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	66	4750	31	2150	0	130	0	65	0	180
X1Y2	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	0	5450	0	2450	0	150	0	75	0	220
X0Y3	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y3	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y4	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y4	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y5	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	0	5350	0	2150	0	130	0	65	0	180
X1Y5	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	0	5450	0	2450	0	150	0	75	0	220
X0Y6	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y6	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220

* Global Clock column represents track count; while other columns represents cell counts

Design Route Status

	# nets
# of logical nets	1451
# of nets not needing routing	573
# of internally routed nets	573
# of routable nets	878
# of fully routed nets	878
# of nets with routing errors	0

Responder:

1. ¿Cuál es el área total del diseño y cuántas celdas contiene?

   1,020 celdas y 6,828.80 um^2

2. ¿Cuál es el período del reloj y qué frecuencia representa?

Clock Summary

Clock	Waveform(ns)	Period(ns)	Frequency(MHz)
clk	{0.000 5.000}	10.000	100.000

3. ¿Qué significan WNS y TNS? ¿El diseño cumple timing?

```
| Design Timing Summary
| ---------------------

    WNS(ns)      TNS(ns)  TNS Failing Endpoints  TNS Total Endpoints      WHS(ns)      THS(ns)  THS Failing Endpoints  THS Total Endpoints     WPWS(ns)     TPWS(ns)  TPWS Failing Endpoints  TPWS Total Endpoints
    -------      -------  ---------------------  -------------------      -------      -------  ---------------------  -------------------     --------     --------  ----------------------  --------------------
    -21.799     -342.102                     17                   19       -0.076       -0.076                      1                   19        4.650        0.000                       0                    55
```

No cumple con el timing debido a la necesidad de processar las divisiones de 16 bits , para las sumas con el prefix adder son cumlidas entre el tiempo necesario antes de un ciclo .

WNS : Holgura temporal del camino más largo del diseño
TNS : Suma de las holguras negativas de cada punto final

4. ¿Cuál es el critical path? Identificar su Startpoint, Endpoint y Path Group.

```
Timing Report

Slack (VIOLATED) :        -21.799ns  (required time - arrival time)
  Source:                 a_reg_reg[15]/C
                            (rising edge-triggered cell FDCE clocked by clk  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:            flag_z_reg/D
                            (rising edge-triggered cell FDCE clocked by clk  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:             clk
  Path Type:              Setup (Max at Slow Process Corner)
  Requirement:            10.000ns  (clk rise@10.000ns - clk rise@0.000ns)
  Data Path Delay:        31.456ns  (logic 8.787ns (27.934%)  route 22.669ns (72.066%))
  Logic Levels:           69  (CARRY4=33 LUT3=2 LUT4=3 LUT5=16 LUT6=15)
  Clock Path Skew:        -0.145ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):    1.707ns = ( 11.707 - 10.000 )
    Source Clock Delay      (SCD):    2.013ns
    Clock Pessimism Removal (CPR):    0.161ns
  Clock Uncertainty:      0.235ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE + UU
    Total System Jitter     (TSJ):    0.071ns
    Total Input Jitter      (TIJ):    0.000ns
    Discrete Jitter          (DJ):    0.000ns
    Phase Error              (PE):    0.000ns
    User Uncertainty         (UU):    0.200ns
```

5. ¿Qué operaciones o celdas provocan el mayor retraso en ese camino?

```
                         0.060    28.635 r  u_alu/u_divisor/cociente0_inferred__12/i__carry__0/CO[3]
net (fo=56, unplaced)    0.603    29.238    u_alu/u_divisor/cociente0_inferred__12/i__carry__0_n_0
LUT5 (Prop_lut5_I1_O)    0.053    29.291 r  u_alu/u_divisor/i__carry_i_2__25/O
net (fo=4, unplaced)     0.364    29.655    u_alu/u_divisor/i__carry_i_2__25_n_0
LUT6 (Prop_lut6_I4_O)    0.053    29.708 r  u_alu/u_divisor/i__carry_i_3/O
net (fo=1, unplaced)     0.363    30.071    u_alu/u_divisor/i__carry_i_3_n_0
CARRY4 (Prop_carry4_DI[1]_CO[3])
                         0.307    30.378 r  u_alu/u_divisor/cociente0_inferred__13/i__carry/CO[3]
net (fo=1, unplaced)     0.008    30.386    u_alu/u_divisor/cociente0_inferred__13/i__carry_n_0
CARRY4 (Prop_carry4_CI_CO[3])
                         0.060    30.446 r  u_alu/u_divisor/cociente0_inferred__13/i__carry__0/CO[3]
net (fo=4, unplaced)     0.542    30.988    u_alu/u_divisor/cociente0_inferred__13/i__carry__0_n_0
LUT5 (Prop_lut5_I4_O)    0.056    31.044 f  u_alu/u_divisor/result[14]_i_3/O
net (fo=15, unplaced)    0.393    31.437    u_alu/u_divisor/result[14]_i_3_n_0
LUT4 (Prop_lut4_I3_O)    0.056    31.493 f  u_alu/u_divisor/result[4]_i_2/O
net (fo=1, unplaced)     0.340    31.833    u_alu/u_divisor/result[4]_i_2_n_0
LUT6 (Prop_lut6_I0_O)    0.053    31.886 f  u_alu/u_divisor/result[4]_i_1/O
net (fo=2, unplaced)     0.351    32.237    u_alu/u_divisor/D[4]
LUT4 (Prop_lut4_I2_O)    0.053    32.290 r  u_alu/u_divisor/flag_z_i_4/O
net (fo=1, unplaced)     0.340    32.630    u_alu/u_divisor/flag_z_i_4_n_0
LUT5 (Prop_lut5_I4_O)    0.053    32.683 r  u_alu/u_divisor/flag_z_i_3/O
net (fo=1, unplaced)     0.340    33.023    u_alu/u_divisor/flag_z_i_3_n_0
LUT5 (Prop_lut5_I2_O)    0.053    33.076 r  u_alu/u_divisor/flag_z_i_2/O
net (fo=1, unplaced)     0.340    33.416    u_alu/u_divisor/flag_z_i_2_n_0
LUT5 (Prop_lut5_I4_O)    0.053    33.469 r  u_alu/u_divisor/flag_z_i_1/O
net (fo=1, unplaced)     0.000    33.469    flag_z_comb
FDCE                                    r  flag_z_reg/D
```

6. ¿Qué optimizaciones implementarían para mejorar el diseño? Justificar su respuesta

Implementar un mejor diseño del divisor ya que el divisor debe de procesar 16 bits en el tiempo que el sumador ha recorrido casi 3 o más ciclos haciéndolo más eficiente que el divisor , el divisor al depender de un módulo de resta debe de procesar en tiempo real las operaciones , estas operaciones generalmente tienen un componente de hardware que esta diseñado para manejar estas operaciones lo cual los hace eficientes comparados a una implementación en normal sin optimización

Para estos casos podemos usar Paralelismo o un Pipeline que funcione de una manera en la que prepara cada proceso antes de realizarse con un fin definido , cada una tiene sus pros y contras pero permite abstraer el hardware con mejores procesos .

2. Generar y analizar los siguientes reportes para ICC2:
    1. Ejecutar las etapas de implementación física:
        1. place_opt → clock_opt → route_auto → route_opt
        2. Generar reportes de QoR y timing después de cada etapa.
    2. Responder:
        1. ¿Cómo cambia el WNS entre las diferentes etapas?

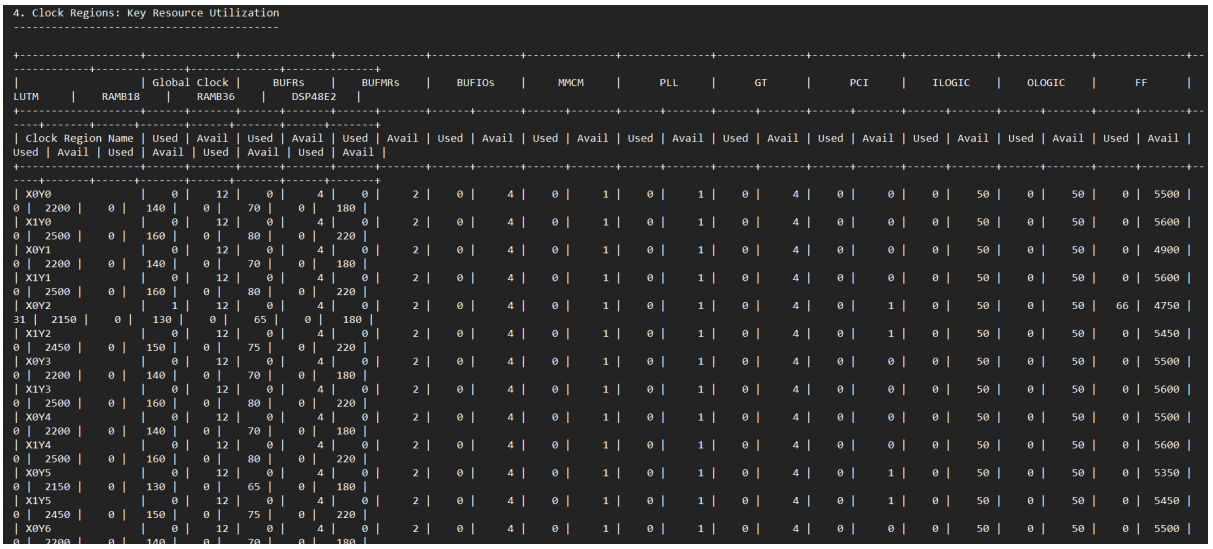
        Mejora y reduce el uso de la holgura generada por el camino critico al mapear directamente los componentes a un diseño que permite optimizar el uso de las compuertas y sus componentes.

2. Slice Logic Distribution

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice	211	0	0	75900	0.28
SLICEL	121	0			
SLICEM	90	0			
LUT as Logic	724	0	0	303600	0.24
using O5 output only	0				
using O6 output only	549				
using O5 and O6	175				
LUT as Memory	0	0	0	130800	0.00
LUT as Distributed RAM	0	0			
using O5 output only	0				
using O6 output only	0				
using O5 and O6	0				
LUT as Shift Register	0	0			
using O5 output only	0				
using O6 output only	0				
using O5 and O6	0				
Slice Registers	55	0	0	607200	<0.01
Register driven from within the Slice	12				
Register driven from outside the Slice	43				
LUT in front of the register is unused	9				
LUT in front of the register is used	34				

2. ¿Cómo afectan el clock skew y las latencias del reloj al timing?

Se eliminan hold times que puedan haber afectado con el diseño y puedan crear ciertos errores directos con el presupuesto de cada reloj.

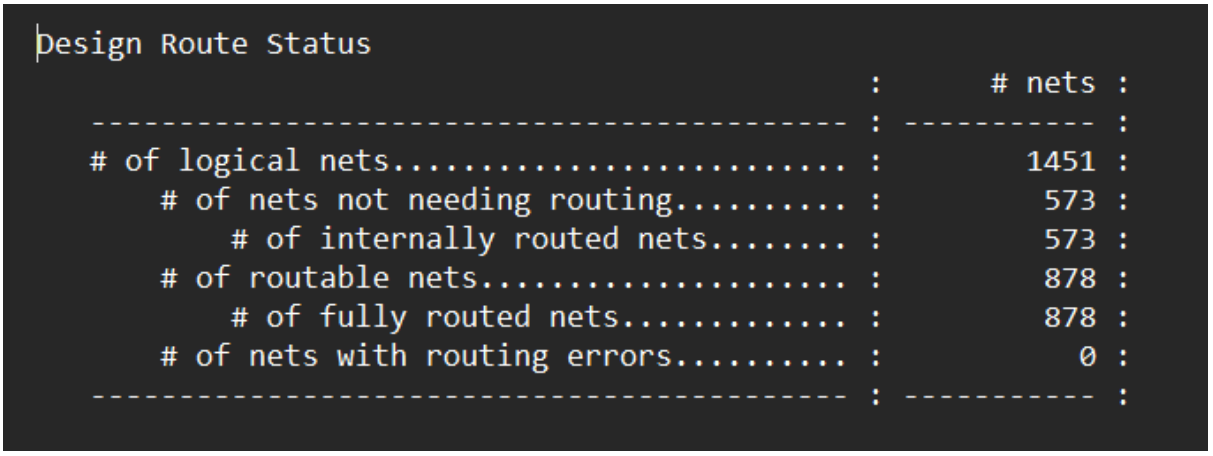


4. Clock Regions: Key Resource Utilization

Clock Region Name	Global Clock		BUFRs		BUFMRs		BUFIOs		MMCM		PLL		GT		PCI		ILOGIC		OLOGIC		FF		LUTM		RAMB18		RAMB36		DSP48E2	
	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail	Used	Avail
X0Y0	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y0	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y1	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	4900	0	2200	0	140	0	70	0	180
X1Y1	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y2	1	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	66	4750	31	2150	0	130	0	65	0	180
X1Y2	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	0	5450	0	2450	0	150	0	75	0	220
X0Y3	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y3	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y4	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180
X1Y4	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5600	0	2500	0	160	0	80	0	220
X0Y5	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	0	5350	0	2150	0	130	0	65	0	180
X1Y5	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	1	0	50	0	50	0	5450	0	2450	0	150	0	75	0	220
X0Y6	0	12	0	4	0	2	0	4	0	1	0	1	0	4	0	0	0	50	0	50	0	5500	0	2200	0	140	0	70	0	180

3. ¿Aumentó el área respecto a Design Compiler? ¿Por qué?

Si ya que debe enrutar las señales en base al diseño comparado con el RTL del design compiler



Design Route Status

	# nets
# of logical nets	1451
# of nets not needing routing	573
# of internally routed nets	573
# of routable nets	878
# of fully routed nets	878
# of nets with routing errors	0

4. ¿Existen violaciones de setup, hold, capacitancia o transición? No

```
2.1 Environment
---------------

+-----------------------+----------------------
| Ambient Temp (C)      | 25.0
| ThetaJA (C/W)         | 1.4
| Airflow (LFM)         | 250
| Heat Sink             | medium (Medium Profil
| ThetaSA (C/W)         | 2.4
| Board Selection       | medium (10"x10")
| # of Board Layers     | 12to15 (12 to 15 Laye
| Board Temperature (C) | 25.0
+-----------------------+----------------------
```

5. ¿Qué muestran las verificaciones de routing, legalidad y conexiones de alimentación?

Las conexiones se vuelven más eficientes con el mapeo directo a hardware

Waveform dentro de Vivado

